Bangladesh University of Business and Technology (BUBT)
Faculty of Engineering& Applied Sciences (FEAS)
Department of Computer Science and Engineering (CSE)

THEORY COURSE OUTLINE

1	**Program**	B.Sc. Engg. in CSE
2	**Course Code**	CSE 207
3	**Course Title**	Database Systems
4	**Course Type**	Core Course
5	**Academic Session**	Fall 2025
6	**Credit Hour**	3.0
7	**Intake**	53
8	**Section**	1
9	**Pre-requisites**	CSE 231- Data Structure
10	**Campus**	Permanent Campus
11	**Course Teacher**	**Name**: Farha Akhter Munmun; **Designation**: Assistant Professor **Specialization**: Networking, Bioinformatics, Machine Learning. **Room No. 310/B1**; **Email:** farha.akter@bubt.edu.bd; **Cell No.** 01707028634

12 Class Schedule

Class Day	Class Hours	Class Room
Monday	2:45 PM – 4.00 PM	905 (B-3)
Tuesday	10:30 AM – 11.45 AM	203 (B-1)

13 Counseling Schedule

Class Day	Class Hours	Class Room
Tuesday	11:45 AM – 1.00 PM	310(B-1)
Wednesday	11:45 AM – 1.00 PM	310(B-1)

14	**Course Objectives**	This course introduces the fundamental concepts and practices of designing and implementing database system with an emphasis on how to organize, maintain and retrieve - efficiently, and effectively - information from a DBMS. It also enables the students to design and perform complex query operations on relational databases. It builds the capability of optimizing the databases efficiently by applying different techniques. It also helps the students to understand the concept of a database, concurrency control.
15	**Course Synopsis**	Introduction to database Concepts: Three level architecture, DBMS, DDBMS, Database administration, models and languages; Relational Algebra; SQL; Intermediate SQL; Database Design: E-R Approach, Relational Model Design, Normalization; File organizations and data structures, Indexing and Hashing, query optimization; Transaction Management: Transaction, Concurrency control.
16	**Text Book**	1. Database System Concept - Silberschatz, Korth, Sudarshan
17	**Reference Book**	1. Database Management System - Raghu Ramakrishna, Johannes Gehrke

		2. Database Systems - Elmasri and Navathe
18	**Course Outcomes (COs)**	Upon completing this course students will be able to: **CO1**: **Understand** the basic concepts of Database management systems (DBMS) and the terms related to database management systems. **CO2: Explain** Structured Query Language (SQL) and relational algebra on relational data model for data manipulation. **CO3**: **Apply** data models (E-R model), normalization techniques to design well defined and accurate database. **CO4: Analyze** the performance of different storage and indexing strategies and concurrency control of transactions.

Mapping of COs to POs

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3			√									
CO4		√										

CO No.	PO No.	Bloom's Domain / Level	Delivery Methods / Activities	Assessment Tools
CO1	PO1	Cognitive / Understanding	Class Lecture	Midterm
CO2	PO1	Cognitive / Understanding	Class Lecture	Midterm
CO3	PO3	Cognitive / Applying	Class Lecture	Final
CO4	PO2	Cognitive/ Analyzing	Class Lecture	Final

19	**Teaching Strategy**	Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. Multimedia projector and a PC will be used for the convenience of the students to understand codes practically. Students must participate in classroom discussions for case studies, problems solving and project developments.
20	**Assessment and Marks Distribution:**	

Class Participation	:	5%
Assignment/Presentation	:	10%
Class Test	:	15%
Midterm Examination	:	30%
Final Examination	:	40%

21 Lecture Plan (Weekly Schedule)

Week	Lecture #	Selected Topics	Chapter #	COs	Assessment
1	1	Introduction: Overview of the course, General discussion about database, applications, drawbacks of traditional file processing system, level of data abstraction, instance, schema, Data model	01	CO1	
	2	Database Languages: DDL, DML; overview of database design, Storage management	01	CO1	
		Query processing, Transaction management, History of database systems, Database Users and Administrators.			
2	3	Introduction to relational model: database, Example of relation	02	CO2	
		Attribute, attribute types			
	4	Relation schema and instance	02	CO2	
		Keys, schema diagram, relational query language			
3	5	Introduction to SQL: Overview of the SQL Query Language; SQL Data Definition: Basic data types, Schema Definition, Integrity constraints	03	CO2	
		Basic Structure of SQL Queries: Queries on Single and Multiple Relations; Cartesian product, Natural Join			**Mid Term Exam** **30**
	6	Additional Basic Operations: rename, string, order operations, where clause predicate; Set operations; aggregate functions	03	CO2	
4	7	Modification of the Database, Nested Subqueries	03	CO2	
	8	Nested Subqueries (Continued)	03 **CT-1**	CO2	
5	9	Intermediate SQL: Join operations: Inner join, Outer join	04	CO2	
	10	View, materialized view, view update	04	CO2	
6	11	Integrity constraints: not null, unique, primary key, check, referential integrity	04	CO2	
		Cascading actions, index creation, user defined domain, authorization			
	12	Formal Relational Query Languages: Relational algebra, select operation, project operation, set operation,	06	CO2	
7	13	Cartesian product, rename, natural join, assignment, outer join, division operation, aggregate functions	06	CO2	
		Join: inner join, outer join using Relational algebra			
	14	Review class for Mid-Term Examination			
8		**Midterm Examination**			
9	15	Entity Relationship Model: Entity set, relationship set, cardinality constraints, participation constraint, ER diagram, Degrees of relationship set, Attribute types	07	CO3	
		Mapping cardinality, Entity Roles, Weak entity set,	07	CO3	

	16	Reduction to relational model, Specialization, Generalization, Aggregation Overview of design, build and test real life database design.	Web		
10	17	Relational Database Design: Atomic attribute, Decomposition, Functional dependency Closure set, Determine keys(Super key, Candidate key, Primary key) from closure set	07 Web **CT-2**	CO3	
	18	Determine canonical cover from Functional dependency using closure set	08	CO3	
11	19	Normalization and Database Design: 1^{st}, 2^{nd} Normalization and Database Design: 3^{rd} Normal form and Boyce-Codd Normal Form (BCNF)	08	CO3	Final Exam 40
	20	Determine Normal Forms, Normalize database, De-normalization	08	CO3	
12	21	Indexing and Hashing: Search key, index file, ordered indices, dense indices, sparse indices, introduction to Hash indices, index evaluation metric	08 Web **CT-3**	CO3	
	22	Introduction to multilevel index: B+ tree, construction and insertion, update, delete in B+ tree	11	CO4	
13	23	Hashing: Hash function, bucket, static hashing, handling of bucket overflow Dynamic hashing, overview of extendible hash structure	11	CO4	
	24	Transactions: State diagram of Transaction, ACID properties of Transaction. Storage Structure, Transaction Atomicity and Durability	14	CO4	
14	25	Concurrency Control of Transactions	14	CO4	
	26	Final Exam Review Class			
15		**Final Exam**			

22 **Overall CO Assessment Criteria**

Assessment methods of COs are given below:

Assessment Area	CO				Assessment Area Mark
	CO1	**CO2**	**CO3**	**CO4**	
Class Participation					
Assignment/Presentation					
Class Test					
Midterm Exam	15	15			30
Final Exam			25	15	40
Total Mark	**15**	**15**	**25**	**15**	**70**

23 **Rubrics**

COs (Bloom's Level)	**Excellent (80%-100%)**	**Good (70%-79%)**	**Satisfactory (60%-69%)**	**Poor (40%-59%)**	**Unsatisfactory (0-39%)**	**Marks (70)**

		CO1 (Understanding)	Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.	Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.	Answer is brief with insufficient detail provided to support issues were introduced.	Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.	None of the relevant details were included or didn’t answer.
		CO2 (Understanding)	Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.	Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.	Answer is brief with insufficient detail provided to support issues were introduced.	Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.	None of the relevant details were included or didn’t answer.
		CO3 (Applying)	The question is answered appropriately by applying the suggested method in the question.	The question is answered briefly by applying the suggested method in the question.	The question is answered correctly by applying the suggested method in the question but some steps are missing.	The question is answered incompletely by applying the suggested method in the question but some steps are correct.	No attempt to implement the suggested method.
		CO4 (Analyzing)	A clear, complete, and properly ordered chain of analyzing steps (i.e. proper explanation of the procedure) is followed to answer the question.	The chain of analyzing steps is complete and correctly ordered but lack of expected explanation.	One or more intermediate analyzing steps are missing or unclear, but the correctness of the analysis is not compromised.	One or more intermediate analyzing steps are missing or unclear to answer the question.	The stated chain of analysis does not lead to the stated question.
24	**Grading Policy**	The following chart will be followed for grading. This has been customized from the guideline provided by the School of Engineering and Computer Science.					

A+	A	A-	B+	B	B-	C+	C	D	F
≥ 80	75-<80	70-<75	65-<70	60-<65	55-<60	50-<55	45-<50	40-<45	<40

25	**Additional Course Policies**	Assignments	There will be at least two assignments. Average marks of the assignments will be counted. No late homework will be accepted. ***Any kind of copy/manipulation in assignment will carry a zero mark.***

			Two or more copied assignments will carry zero mark in all assignments. **Zero tolerance will be shown in this regard.** Solutions to assignment problems will be provided through web and on hand.
		Class Test	There will be at least three class tests (CT). Best two of three or best three of four CTs will be counted. Both of regular and surprise CTs can be conducted.
		Exams	CT, Mid-term and final exam will be closed book, closed notes. Mobile phones are strictly prohibited in exam halls. Students are insisted to carry their own watch and synchronize time during exam hours.
		Test Policy	If a student is absent from class test anyway and makes no report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed as 2 of 3 or 3 of 4 CTs are being considered. No make-up for Mid-exam will be entertained without physical presence and recommendation of the guardian along with written permission of the department. Make-up of Mid-exam may be much harder than the regular one.
26	**Additional Information**	a. Academic Calendar Summer 2020: http://www.bubt.edu.bd/academics/academic-calendar. b. Academic Policies: http://www.bubt.edu.bd/academics/academic-rules-a-regulations. c. Grading & Evaluation: http://www.bubt.edu.bd/academics/academic-rules-a-regulations. d. Proctorial Rules: http://www.bubt.edu.bd/administrator/proctors-office.	

27 Bloom's Taxonomy for Teaching-Learning

Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specificity. The three lists cover the learning objectives in Cognitive, Affective and Psychomotor domains. The Cognitive domain list has been the primary focus of most education and is frequently used to structure curriculum learning objectives, assessments and activities. The three domains and respective levels are illustrated below.

Cognitive [C] (Knowledge-based)	**Affective [A]** (Emotion-based)	**Psychomotor [P]** (Action-based)
1. Remembering	1. Receiving	1. Imitating
2. Understanding	2. Responding	2. Manipulating
3. Applying	3. Valuing	3. Précising
4. Analyzing	4. Organizing	4. Articulating
5. Evaluating	5. Characterizing	5. Naturalizing
6. Creating	--- --- ---	--- --- ---

28 Descriptions of Cognitive Domain (Anderson and Krathwohl's Taxonomy 2001):
The **cognitive domain** involves the development of our mental skills and the acquisition of knowledge.

Level	**Category**	**Meaning**	**Keywords**
C1	Remembering	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, write
C2	Understanding	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, report

	C3	Applying	Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, use
	C4	Analyzing	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, structure
	C5	Evaluating	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, critique, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor
	C6	Creating	Putting elements together to form a coherent or functional whole ;reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function.	Construct, design, develop, generate, hypothesize, invent, plan, produce, compose, create, make, perform, plan, produce

29 Graduate Attributes (Program Outcomes) for B.Sc. in Engineering Program based on Washington Accord

Program Outcomes (POs) are narrower statements that describe what students are expected to know and be able to do by the Time of graduation. These relate to the knowledge skills and attitudes that students acquire while progressing through the program. The students of the B.Sc. in EEE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation.

PO1–Engineering knowledge (Cognitive): Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2–Problem analysis (Cognitive): Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.

PO3–Design/development of solutions (Cognitive, Affective): Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.

PO4–Investigation (Cognitive, Psychomotor): Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

PO5–Modern tool usage (Psychomotor, Cognitive): Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6–The engineer and society (Affective): Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7–Environment and sustainability (Affective, Cognitive): Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8–Ethics (Affective): Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.

PO9–Individual work and teamwork (Psychomotor, Affective): Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.

PO10–Communication (Psychomotor, Affective): Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.

PO11–Project management and finance (Cognitive, Psychomotor): Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects

	in multidisciplinary environments. **PO12–Life-long learning (Affective, Psychomotor):** Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.
30	**Social & Moral Capital**
	Our promises are based on the three cardinal principles: (a) What we do believe (b) What we do practice, and (c) What we will promote However, students are advised to undertake the following commitments for moral development.

1. To be punctual and attentive in class **2.** To maintain inclusive learning environment **3.** To ensure mutual respect **4.** To be cooperative in group learning. **5.** To be innovative and Creative **6.** To follow dress code and wearing ID card **7.** To be always proactive	**8.** Try to follow and review day to day class **9.** To avoid conspiracy **10.** To prioritize honesty & faith **11.** To be motivated for asking question and encourage feedback **12.** To develop attitude for speaking in English **13.** Do not ignore to carry out any assignments or commitments **14.** To be clean and decent in all levels.	**15.** To be sincere for class preparation **16.** Do not forget to switch-off the cell phone in class **17.** Do not forget to carry course pack and learning stuffs in class **18.** To maintain loyalty and trust to the university **19.** Must avoid unfair means and plagiarism in exam, reports and assignments **20.** Must maintain eco-friendly environment in the campus.

Farha Akhter Munmun,
Assistant Professor, Dept. of CSE, BUBT.

Prepared by: **Checked by:** **Approved by:**